Multimodal Artificial Intelligence: Foundations and Frontiers

Page 1: Introduction and Vision-Language Architectures

1. Executive Overview:
Multimodal AI refers to models capable of processing and relating information from multiple modalities, including text, computer vision, audio, and structured sensor streams. The unifying objective is to build cross-modal representations where semantic concepts are shared across modalities.

2. Contrastive Pretraining:
Pioneered by OpenAI's CLIP (Contrastive Language-Image Pre-Training), contrastive architectures train an image encoder (e.g., Vision Transformer ViT) and a text encoder simultaneously on web-scale pairs.
The model maximizes the cosine similarity between matched (image, text) embeddings while minimizing similarity
between mismatched negative pairs in the batch using a symmetric cross-entropy loss.

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Page 2: Autoregressive Vision-Language Models and Cross-Attention

3. Cross-Attention and Perceiver Resamplers:
Modern Vision-Language Models (VLMs) such as Gemini, LLaVA, and Flamingo project visual patches into the token embedding space of large language models. Cross-attention layers enable the language decoder to attend directly to visual features during autoregressive token generation.

4. Benchmark Evaluations:
State-of-the-art multimodal systems are evaluated on rigorous benchmarks such as:
- MMLU (Massive Multitask Language Understanding) for academic and reasoning performance.
- MMMU (Massive Multi-discipline Multimodal Understanding) assessing university-level visual domain expertise.
- MathVista for multimodal mathematical reasoning.

5. Safety and Alignment:
Multimodal models introduce novel attack vectors, such as typographic visual prompt injections, where adversarial
text rendered inside images attempts to override safety boundaries. Robust alignment requires multimodal RLHF.